

Satellite1 Ultra

Build manual for a serviceable passive-radiator enclosure for the FutureProofHomes Satellite1 development kit.

Source commit **e3840a5a3158**. Units: millimetres.

Status: IN DEVELOPMENT. No physical validation has been performed. Every geometric statement in this manual is machine-checked against the authoritative CAD. Every acoustic number is a lumped-parameter simulation, not a measurement. Nothing here may be quoted as measured fit, acoustic, thermal, wireless or wake-word performance.

1. Product envelope

Property	Value	Evidence
Cabinet section	160 x 180 mm	VERIFIED_DIGITALLY
Overall shell section	192 x 212 mm	VERIFIED_DIGITALLY
Overall height with official top	237.7 mm	VERIFIED_DIGITALLY
Net acoustic volume	3.447 L	VERIFIED_DIGITALLY
Assembled mass	3488 g	ENGINEERING_ESTIMATE
Minimum tipping angle	49.6 deg	ENGINEERING_ESTIMATE



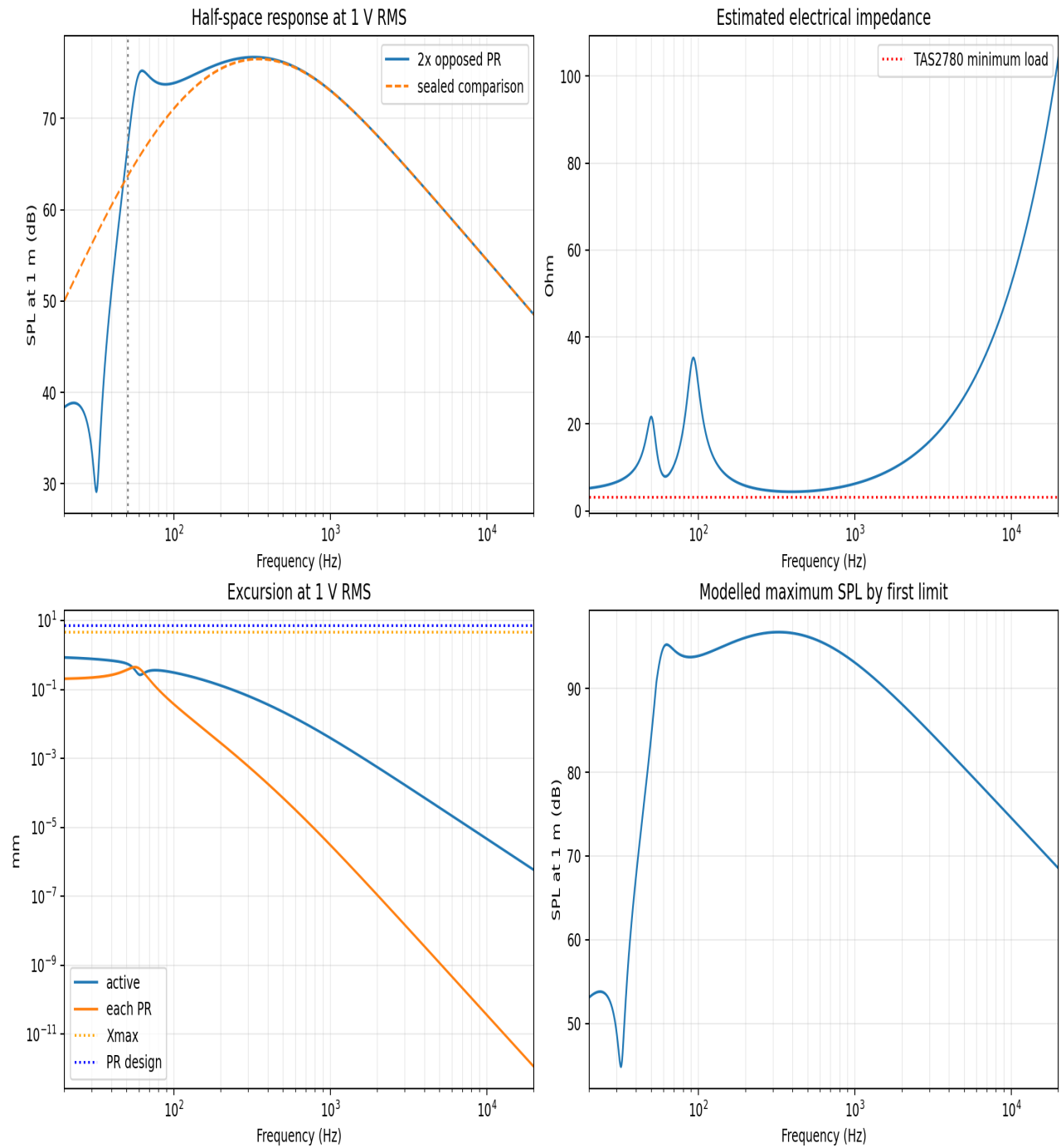
2. Acoustic design

Quantity	Modelled value
Architecture	1 active driver, 2 opposed passive radiators
Net volume	3.447 L
System tuning	60.0 Hz
Added mass per radiator	1.07 g
f3, passive-radiator alignment	56.9 Hz

Quantity	Modelled value
f3, same driver sealed	132.1 Hz
Minimum impedance	4.39 ohm
Maximum SPL at 100 Hz	93.9 dB at 1 m
Protective high-pass	51.0 Hz, order 4

ENGINEERING_ESTIMATE. These are calculated from published manufacturer parameters and the exact CAD net volume. They are not measurements. Measure impedance first and feed the result back into the model before finalising any DSP.

Satellite1 Ultra lumped-parameter simulation — 3.447 L net — NOT A MEASUREMENT



3. Bill of materials

item	category	quantity	material	source	evidence
anti_slip_ring	printed part	1	TPU 95A	this repository, exports/step	VERIFIED_DIGITALLY
outer_shell	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
main_cabinet	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
pressure_divider	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
electronics_shroud	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
active_driver_clamp_ring	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
passive_radiator_clamp_ring	printed part	2	ASA	this repository, exports/step	VERIFIED_DIGITALLY
divider_gasket	printed part	1	2 mm closed-cell EPDM	this repository, exports/step	VERIFIED_DIGITALLY
driver_gasket	printed part	1	2 mm closed-cell EPDM	this repository, exports/step	VERIFIED_DIGITALLY
passive_radiator_gasket	printed part	2	2 mm closed-cell EPDM	this repository, exports/step	VERIFIED_DIGITALLY
cable_gland	printed part	1	TPU 95A	this repository, exports/step	VERIFIED_DIGITALLY
base_skirt	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
bottom_service_plate	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
ballast_cartridge	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
ballast_cartridge_lid	printed part	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_official_interface	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_active_driver	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_passive_radiator	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_heat_set_insert	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_gasket_base	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_gasket_cap	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
coupon_cable_passage	fit coupon	1	ASA	this repository, exports/step	VERIFIED_DIGITALLY
Dayton Audio ND91-4	acoustic component	1	-	references/manufacturer/Dayton_ND91-4-2002-Manufacturer-Datasheet.pdf	DERIVED_FROM_MANUFACTURER_DATA
SB Acoustics SB12PACR-00	acoustic component	2	-	references/manufacturer/SB_Acoustics_SB12PACR-00-Manufacturer-Datasheet.pdf	DERIVED_FROM_MANUFACTURER_DATA
FutureProofHomes Satellite1 development kit (Official Hardware)	Official Hardware	1	-	reference-assets/MANIFEST.csv	DERIVED_FROM_OFFICIAL_CAD
Official Squirrelle upper stack: mid-plate, threads, top plate, PCB spacer, lock ring	Official Hardware	1	ASA or PETG	reference-assets/official/Satellite1-End-user-manual.pdf	DERIVED_FROM_OFFICIAL_CAD
M3 x 0.5 A2 socket/button head screws, assorted lengths per schedule	hardware	40	A2 stainless	reports/validation/fasteners.json	VERIFIED_DIGITALLY
M3 heat-set inserts Oe4.6 x 5.7 mm	hardware	40	brass	reports/validation/fasteners.json	VERIFIED_DIGITALLY
2.0 mm closed-cell EPDM sheet	consumable	1	EPDM	cut from exports/step gasket profiles	VERIFIED_DIGITALLY
Mild-steel ballast plates 110 x 122 x 5 mm	ballast	2	mild steel	cut to size	ENGINEERING_ESTIMATE
Speaker cable, 2 core, 1.0 mm2, 350 mm	consumable	1	copper/PVC	-	ENGINEERING_ESTIMATE
Added radiator tuning mass, M6 hardware, per radiator	acoustic tuning	2	steel	reports/acoustics/summary.json	ENGINEERING_ESTIMATE

4. Fastener schedule

Joint	Qty	Screw	Stack	Engage	Margin	Access
official mid-plate to pressure divider	4	M3x6	1.0	5.0	2.2	top, through the official Ø6.5 counterbore
electronics shroud to pressure divider	4	M3x8	3.0	5.0	2.2	top
pressure divider to cabinet	8	M3x10	5.5	4.5	2.7	top
active-driver clamp ring to cabinet	4	M3x10	5.0	5.0	2.2	front (-Y), grille cage removed
passive-radiator clamp ring to cabinet (each)	4	M3x10	5.0	5.0	2.2	side (+/-X), grille cage removed
base skirt to cabinet	4	M3x10	5.0	5.0	2.2	bottom, service plate removed
bottom service plate to base skirt	4	M3x8	4.0	4.0	3.2	bottom
grille cage to bottom service plate	4	M3x8	4.0	4.0	3.2	bottom

Every insert bore is blind and drilled deeper than the insert, so no screw can bottom. No fastener crosses the acoustic pressure boundary.

5. Sealing

- integral cabinet walls, acoustic floor and component mounting pads
- compressed 2 mm EPDM divider gasket on a continuous 4 mm rim land
- compressed 2 mm EPDM active-driver gasket on a continuous seat land
- two compressed 2 mm EPDM radiator gaskets on continuous seat lands
- TPU interference gland around the speaker-wire pair
- all cabinet fasteners are blind heat-set inserts; none crosses the boundary

Sealing gate status: **PASS**. The gate proves, on the B-rep, that every gasket land is a continuous annulus and that no fastener bore breaks through it. It cannot prove gas tightness of a printed wall; that is `REQUIRES_PHYSICAL_VALIDATION`.

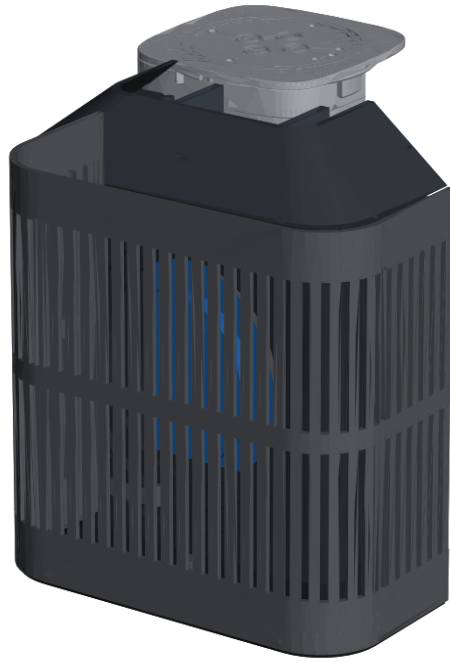
6. Assembly

1. install all heat-set inserts
2. load the steel ballast, close the cartridge lid
3. fit the active-driver gasket and driver, install the clamp ring
4. fit both radiator gaskets and radiators, install both clamp rings
5. route the speaker pair through the divider and fit the TPU gland
6. fit the divider gasket and bolt the pressure divider to the cabinet
7. bolt the base skirt up into the acoustic floor
8. lower the outer shell over the cabinet
9. seat the ballast cartridge and close the bottom service plate
10. connect the boards and bolt the electronics shroud to the divider
11. invert and bolt the outer shell to the bottom service plate
12. seat the official mid-plate and upper stack on the divider bosses
13. stretch the TPU anti-slip ring onto the base



7. Cross sections





8. Manufactured parts

Satellite1 Ultra — manufactured parts in print orientation

anti_slip_ring	outer_shell	main_cabinet	pressure_divider	electronics_shroud
active_driver_clamp_ring	passive_radiator_clamp_ring	divider_gasket	driver_gasket	passive_radiator_gasket
cable_gland	base_skirt	bottom_service_plate	ballast_cartridge	ballast_cartridge_lid
coupon_official_interface	coupon_active_driver	coupon_passive_radiator	coupon_heat_set_insert	coupon_gasket_base
coupon_gasket_cap	coupon_cable_passage			

9. Validation gate summary

Gate	Status	Evidence
acoustic_volume	PASS	VERIFIED_DIGITALLY
assembly	PASS	VERIFIED_DIGITALLY
center_of_gravity	PASS	ENGINEERING_ESTIMATE
clearance	PASS	VERIFIED_DIGITALLY
collision	PASS	VERIFIED_DIGITALLY
core_fit	PASS	VERIFIED_DIGITALLY
fasteners	PASS	VERIFIED_DIGITALLY
printability	PASS	VERIFIED_DIGITALLY
sealing	PASS	VERIFIED_DIGITALLY
tolerance	PASS	ENGINEERING_ESTIMATE
wall_thickness	PASS	VERIFIED_DIGITALLY

10. Risk register

ID	Sev	Risk	Mitigation
R-01	HIGH	Component flange thicknesses are not published	coupon_active_driver and coupon_passive_radiator measure the assembled stack directly
R-02	HIGH	Core board position in the official stack is undetermined	The enclosure is validated against a Core-sized free volume in the electronics bay instead
R-03	HIGH	Bass extension is limited by the driver, not the enclosure	Larger-displacement 4 in driver is the single highest-leverage improvement available. Even
R-04	MEDIUM	No pressure-decay measurement exists	Pressure-decay test the assembled cabinet and enter the measured leakage Q into config
R-05	MEDIUM	Printed dimensional accuracy is assumed, not measured	Shred-coupon set measures it; corrections go into config/physical_compensation.yaml
R-06	MEDIUM	Wi-Fi and wake-word performance are unmeasured	Run the documented wake-word and wireless test procedures on an assembled unit again
R-07	MEDIUM	Board thermal behaviour in a closed shroud is unmeasured	Re-run the documented thermal soak procedure; if margins are short, the vent bank is a para
R-08	LOW	Heat-set insert pull-out strength is assumed	coupon_heat_set_insert brackets four bore diameters; pull-test to 250 N.
R-09	LOW	Batch 2 hardware is not supported	The board revision is a configuration value and the official Batch 2 geometry is already pre

11. Before you build

Print the eight fit coupons, measure them against docs/fit-coupons.md, and enter the corrections in config/physical_compensation.yaml. Then regenerate every part. Do not print the full set first.

This design is not PHYSICALLY_VALIDATED. Physical fit, sealing, acoustic output, wake-word behaviour, thermal margin and wireless performance all remain unmeasured.